

Object Oriented Programming

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Last Week

- Objects and classes

This Week

- Object and classes
 - Default constructor
 - Copy constructor
 - Out of class function definitions
 - .h and .cpp file for a class
 - User defined String class
 - Buffer management
 - Extra functionality
 - STL **string** class

Constructor types

- Default constructor
 - `Point(int x1, int y1) { x = x1; y = y1; }`
- Copy constructor
 - `Point(const Point &p2) {x = p2.x; y = p2.y; }`
- `Point p1(10, 15);`
- `Point p2 = p1;`

Distance Class

- Class diagram Distance which holds distance in feet and inches. Member functions set and clear data. Show the use of Constructor and destructor for this class. Add member function to get distance as input from user. Add member function to add feet (add_feet) and inches (add_inches). Add member function “add_distance” to add distance provided as an argument.

C Strings

- Arrays of characters
- Null character
- Space needs to be allocated

C string buffer management

- Setw to manipulate string size
 - `cin >> setw(MAX) >> str;`
- Strings with spaces
 - `cin.get(str, MAX);`
- Multi line strings
 - `cin.get(str, MAX, '$');`

Quiz

- User defined String class

STL string class

- Compare the two
- string added functionality
- And much more in STL

Summary

- Overloading constructors is a great programming convenience
- Class function members can be defined outside class and even in separate files
- Always try to use classes/functions already available in (STL) library

Next Week

- Objects and classes Contd